

Alexander Simpson

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Education

University of California, Berkeley
Bachelor of Arts in Computer Science

May 2028

Relevant Coursework: CS61A, CS61B, CS70, Data 8, EECS 16A

In Progress (Fall 2025): EECS 126, EECS 16B, CS 61C, Data 100

Skills

Languages: Java, Python, SQL, Scheme

Tools & Technologies: Git, Linux, Jira, NumPy, Pandas, Matplotlib, CuPy, VisPy

Concepts: Simulation modeling, data analysis, parallelization, Agile workflow, debugging/testing

Other: Technical tutoring, leadership, team coordination

Clearance: Previously held TS/SCI with Polygraph; eligible for reactivation

Experience & Technical Projects

Student Computer Intern, National Security Agency – Fort Meade, MD August 2023 – May 2024

- Held Top Secret clearance with polygraph; operated in secure, mission-critical environments.
- Developed and maintained internal tools with software engineering teams, improving the efficiency of cybersecurity workflows by automating manual processes.
- Used Jira in an Agile workflow to track tickets, manage issues, and coordinate fixes across multiple teams
- Supported migration of a secure internal database, performing data integrity validation, and assisting with transition testing

Project: Orbital Mechanics Simulator (Python, NumPy, CuPy, VisPy) Summer 2025

- Built a **real-time orbital physics engine** simulating N-body gravitational interactions with Verlet integration.
- Validated orbital stability over long runs using energy conservation checks, ensuring scientific accuracy.
- Achieved ~40% performance improvement by parallelizing gravitational force calculations with CuPy GPU acceleration
- Integrated **VisPy** for high-performance, interactive 3D visualizations including real-time trajectories, velocity vectors, and system-wide starfield rendering.
- Structured the codebase modularly to support scaling from **two-body interactions** up to **multi-planet**
- Positioned as a foundation for aerospace simulation and trajectory planning, demonstrating scalability from classroom use to real-world modeling

Project: Sickle Cell Allele Simulation (Java, Swing, Concurrency, Matplotlib for visualization) Spring 2025

- Built a **multi-generational genetic simulation** engine modeling sickle cell allele frequency under varying malaria prevalence and healthcare conditions.
- Designed **object-oriented architecture** with Individual genotype objects and encapsulated survival logic (AA, AS, SS).
- Implemented **parallelized mortality and allele updates** using Java concurrency, improving runtime scalability on larger simulated populations.
- Developed a **Java Swing GUI** with real-time parameter controls and visualizations, including allele frequency graphs and family tree panels.
- Applied simulation modeling and data visualization techniques to communicate population genetics dynamics for research and education.

Computer Science Tutor – UC Berkeley

Jan 2024-Present

- Tutored students in **Python, Java, and core CS concepts**, reinforcing data structures, algorithms, and programming fundamentals.
- Helped students debug assignments, understand error messages, and improve problem-solving strategies.
- Guided students on best coding practices, debugging strategies, and effective Git workflows.